

TARA ANGLICAN SCHOOL FOR GIRLS



2003 YEAR 12 EXTENSION ONE MATHEMATICS

HALF YEARLY EXAMINATION

Weight: 25%

Total Marks: 70 marks

Time Allowed: 1.5 Hours + 5 minutes reading time

DIRECTIONS TO CANDIDATES

- There are FIVE (5) questions
- Attempt all questions
- Board Approved calculators may be used
- Start a NEW BOOKLET for each question
- All necessary working should be shown in every question. Marks may be deducted for carelessly or badly arranged work.
- An Integral sheet is provided with this paper

(a) If $\sin A = \frac{1}{\sqrt{7}}$, and $\cos A > 0$, find the exact value of $\sin 2A$.

3

(b) Solve the inequality $\frac{4x+3}{x-4} \geq 1$

3

(c) Deduce, to the nearest minute, the obtuse angle between the lines $\frac{x}{7} + \frac{y}{5} = 1$ and $2x - 3y + 4 = 0$.

3

(d) Use the substitution $u = 1 + t$, to find $\int \frac{t}{\sqrt{1+t}} dt$

3

Question Two (13 Marks) START A NEW BOOKLET

Marks

(a) Integrate the following:

(i) $\int_{-1}^{\frac{1}{2}} \frac{x^3 - 4x}{x} dx$

2

(ii) $\int (4-y)^3 dy$

1.5

(b) (i) Express $7\cos\theta - \sin\theta$, in the form $R\cos(\theta + \alpha)$, where $R > 0$ and $0^\circ \leq \alpha \leq 90^\circ$

2

(ii) Hence, solve $7\cos\theta - \sin\theta = 5$ for $0^\circ \leq \theta \leq 360^\circ$, giving answers to the nearest minute.

2.5

(c) Given $\int_1^x x\sqrt{x} = \frac{62}{5}$, deduce the value of K .

2

(d) Prove $\frac{\sin 2\beta + \sin \beta}{1 + \cos 2\beta + \cos \beta} = \tan \beta$

3

Question Three (19 Marks)

START A NEW BOOKLET

Marks

(a) Given $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are points on the parabola $x^2 = 4ay$,

(i) Derive the equation of the tangent to the parabola at P.

2

(ii) The tangent at P and the line through Q parallel to the y-axis, intersect at T. Determine the co-ordinates of T.

2

(iii) Calculate the co-ordinates of M, the midpoint of PT.

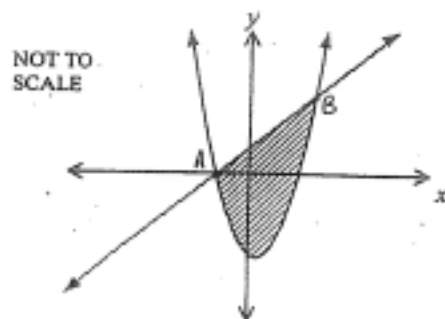
1

(iv) Given $pq = -1$, find the equation of the locus of M.

3

(b) Prove by Mathematical Induction, that $7^n + 11^n$ is divisible by 9, if n is odd.

4

(c) The diagram below shows the curves $y = x^2 - x - 6$ and $y = x + 2$.

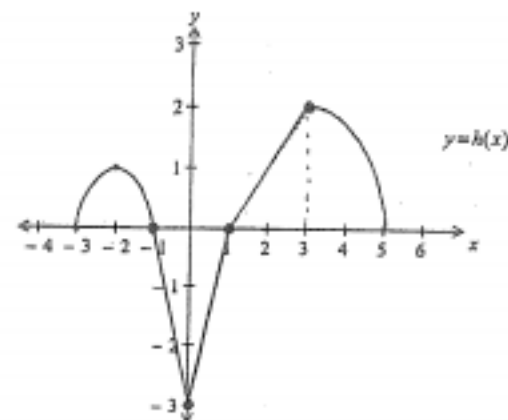
(i) Find the x-co-ordinates of points A and B.

3

(ii) Hence, calculate the area enclosed by the two curves.

3

NOT TO SCALE

(i) Evaluate $\int_{-3}^5 h(x) dx$

2

(ii) Calculate the area enclosed by the curve $y = h(x)$, and the x-axis.

2

(b) (i) Differentiate $y = \frac{2x^2 - 1}{3x^2 + 4}$ (ii) Hence integrate $\int_0^2 \frac{x dx}{(3x^2 + 4)^2}$

1

(c) $P(x)$ is a monic polynomial of degree 4 and has exactly 2 real zeros, at 1 and -1.(i) If $P(x)$ is an even function find a general equation to represent this information.

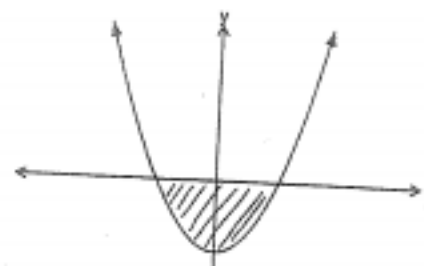
2

(ii) Hence, if $P(x) = 33$ when $x = -2$, find the unique polynomial $P(x)$.

2

(d) The region bounded between $y = x^2 - 1$ and the x-axis is rotated about the x-axis. Determine the exact volume of the solid of revolution formed.

3

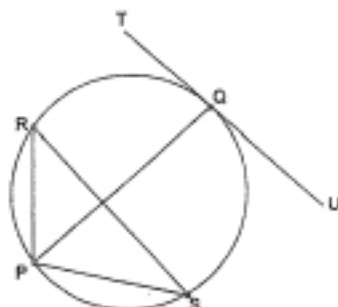


Question Five (11 Marks) START A NEW BOOKLET

Marks

- (a) The diagram below, shows a circle with a chord PQ and another chord RS, which is parallel to the tangent at Q.

4



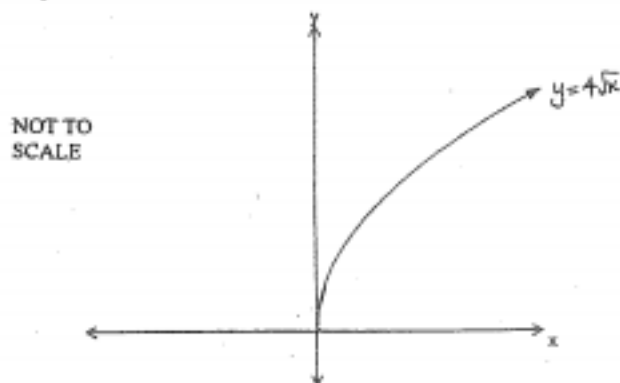
NOT TO SCALE

Copy or trace the diagram onto your page

Prove that chord PQ bisects $\angle RPS$. [HINT: construction lines may be required]

- (b) A vase is formed by the rotation of the curve $y = 4\sqrt{x}$, about the y -axis. Calculate the amount of water needed to fill the vase to a depth of 8 cm.

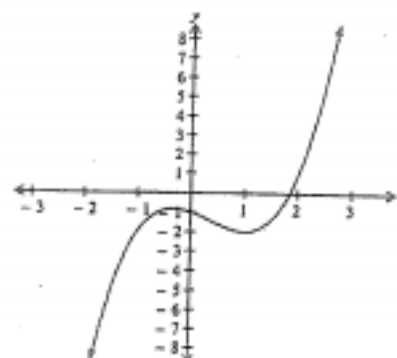
3



Question Five continued...

- (c) The function $f(x) = x^3 - x^2 - x - 1$ is shown below.

NOT TO SCALE



- (i) Using $x = 2$ as a first approximation for $f(x) = 0$, use one application of Newton's method to find a better approximation to 1 decimal place.

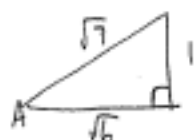
2

- (ii) Copy or trace the diagram. Describe, in words using your diagram, why $x = 1$ is an unsuitable first approximation to this $f(x) = 0$.

2

Q1.1

(a) $\sin A = \frac{1}{\sqrt{5}}$ $\cos A > 0$



$$\begin{aligned}\sin 2A &= 2 \sin A \cos A \\ &= 2 \left(\frac{1}{\sqrt{5}} \right) \left(\frac{\sqrt{4}}{\sqrt{5}} \right) \\ &= \frac{2\sqrt{4}}{5}\end{aligned}$$

(b) $\frac{4x+3}{x-4} \geq 1$ $x-4 \neq 0$
 $\therefore x \neq 4$

x both sides by $(x-4)^{-1}$

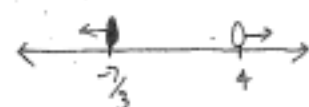
$$(x-4)^{-1} \times \frac{4x+3}{x-4} \geq 1 \times (x-4)^{-1}$$

$$(x-4)(4x+3) \geq x^2 - 8x + 16$$

$$4x^2 - 13x - 12 \geq x^2 - 8x + 16$$

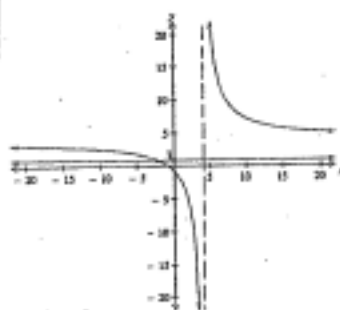
$$3x^2 - 5x - 28 \geq 0$$

$$(3x+7)(x-4) \geq 0$$



test $x=0$ $(1)(-4) > 0$
 $-28 > 0$

$$\therefore x \leq -\frac{7}{3}, x > 4$$



$$y = \frac{x-4}{4x+3}$$

(c) $\frac{x}{7} + \frac{y}{5} = 1$

$$\frac{y}{5} = 1 - \frac{x}{7}$$

$$y = 5 - \frac{5x}{7}$$

$$m_1 = -\frac{5}{7}$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

for obtuse angle,
do not take
absolute value

$$\tan \theta = \frac{-\frac{5}{7} - \frac{3}{5}}{1 + (-\frac{5}{7})(\frac{3}{5})}$$

$$= -\frac{29}{11}$$

$$\theta = 180^\circ - 69^\circ 14'$$

$$\therefore \theta = 110^\circ 46'$$

(d) $u = 1+t$

$$\frac{du}{dt} = 1$$

$$\therefore dt = du$$

$$\int \frac{t}{\sqrt{1+t}} dt = \int \frac{u-1}{\sqrt{u}} du$$

$$= \int u^{\frac{1}{2}} - u^{-\frac{1}{2}} du$$

$$= \frac{2u^{\frac{3}{2}}}{\frac{3}{2}} - 2u^{\frac{1}{2}} + C$$

$$= \frac{2}{3} \sqrt{1+t}^3 - 2\sqrt{1+t} + C$$

Q1.2

(a) (i) $\int_{-1}^k \frac{x^2 - 4x}{x} dx$

$$= \int_{-1}^k x^2 - 4 dx$$

$$= \left[\frac{x^3}{3} - 4x \right]_{-1}^k$$

$$= \left(\frac{1}{24} - 4 \times \frac{1}{2} \right) - \left(-\frac{1}{3} + 4 \right)$$

$$= -5\frac{5}{8}$$

(ii) $\int (4-y)^3 dy = \frac{(4-y)^4}{-4} + C$

(b) (i) $\int_a^b \cos \theta - \sin \theta = 5\sqrt{2} \cos(\theta + 8^\circ 8')$

$$R = \sqrt{7^2 + 1^2}$$

(ii) $5\sqrt{2} \cos(\theta + 8^\circ 8') = 5$

$$\cos(\theta + 8^\circ 8') = \frac{1}{\sqrt{2}}$$

$$\theta + 8^\circ 8' = 45^\circ, 315^\circ$$

$$\theta = 36^\circ 52', 306^\circ 52'$$

(c) $\int_1^k x\sqrt{x} dx = \frac{62}{5}$

$$\int_1^k x^{\frac{3}{2}} dx = \frac{62}{5}$$

$$\left[\frac{2x^{\frac{5}{2}}}{\frac{5}{2}} \right]_1^k = \frac{62}{5}$$

$$\frac{2k^{\frac{5}{2}}}{5} - \frac{2}{5} = \frac{62}{5}$$

$$\frac{2k^{\frac{5}{2}}}{5} = \frac{64}{5}$$

$$k^{\frac{5}{2}} = 32$$

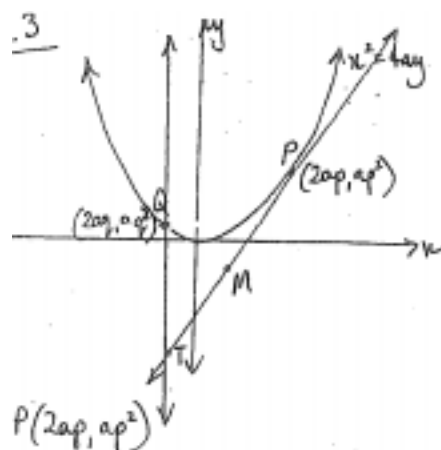
(d) LHS = $\frac{\sin 2\beta + \sin \beta}{1 + \cos 2\beta + \cos \beta}$

$$= \frac{2 \sin \beta \cos \beta + \sin \beta}{1 + (2 \cos^2 \beta - 1) + \cos \beta}$$

$$= \frac{\sin \beta (2 \cos \beta + 1)}{\cos \beta (2 \cos \beta + 1)}$$

$$= \tan \beta$$

$$= \text{RHS.}$$



$$\begin{aligned}
 &= 4ay \\
 &= \frac{x^2}{4a} \\
 &= \frac{x}{2a} \\
 &\therefore \frac{dy}{dx} = \frac{2ap}{2a} = p \\
 &ap^2 = p(x - 2ap) \\
 &ap^2 = px - 2ap^2 \\
 &0 = px - y - ap^2
 \end{aligned}$$

(ii) $x = 2aq$, $0 = px - y - ap^2$ ②

Sub ① into ②

$$\begin{aligned}
 0 &= p(2aq) - y - ap^2 \\
 0 &= 2apq - y - ap^2 \\
 \therefore y &= 2apq - ap^2
 \end{aligned}$$

co-ords of T:

$$(2aq, 2apq - ap^2)$$

(iii) $x = \frac{2ap + 2aq}{2}$

$$\therefore x = a(p+q)$$

$$y = \frac{ap^2 + 2apq - ap^2}{2}$$

$$\therefore y = apq$$

M $(a(p+q), apq)$

(iv) $x = a(p+q)$

$$y = apq \text{ where } pq = -1 \text{ (given)}$$

$$\therefore y = -a.$$

the locus of M is the directrix

(b) $7^n + 11^n$ divisible by 9, n is odd

let $n=1$,

$$7^1 + 11^1 = 18 = 9 \times 2$$

$$\therefore \text{div by } 9.$$

let $n=3$,

$$7^3 + 11^3 = 1674 = 9(186)$$

$$\therefore \text{div. by } 9.$$

Assume true for $n=k$,

$$7^k + 11^k = 9M \text{ where } M \text{ is an integer}$$

Prove true for $n=k+2$, since n is o

$$7^{k+2} + 11^{k+2} = 9P \text{ where } P \text{ is integer}$$

$$LHS = 7^{k+2} + 11^{k+2}$$

cont'd...

$$\begin{aligned}
 &7^2 \cdot 7^k + 11^2 \cdot 11^k \\
 &7^2(9M - 11^k) + 11^2 \cdot 11^k \\
 &\text{where } 7^k + 11^k = 9M \\
 &\text{from assumption}
 \end{aligned}$$

$$\begin{aligned}
 &9 \times 7^2 M - 7^2 \cdot 11^k + 11^2 \cdot 11^k \\
 &= 9 \times 49M + 72 \cdot 11^k \\
 &= 9(49M + 8 \cdot 11^k) \\
 &= 9P \text{ where } P \text{ is an integer,} \\
 &\text{as required.}
 \end{aligned}$$

cc true for $n=1$ and 3 and proved true for k and $n=k+2$, true
all values of n ,

(c) $x^2 - x - 6 = x + 2$

$$x^2 - 2x - 8 = 0$$

$$(x-4)(x+2) = 0$$

$$\therefore x = -2, 4.$$

at A, $x = -2$

at B, $x = 4.$

(ii) $A = \int_{-2}^4 x+2 - (x^2 - x - 6) dx$

$$= \int_{-2}^4 -x^2 + 2x + 8 dx$$

$$= \left[-\frac{x^3}{3} + x^2 + 8x \right]_{-2}^4$$

$$= \left(-\frac{64}{3} + 16 + 32 \right) - \left(\frac{8}{3} + 4 - 16 \right)$$

$$= 2$$

Que 4

(a) (i) $\int_{-3}^5 h(x) dx = \left(\frac{\pi x^4}{2} \right) - \left(\frac{1}{2} x^2 \right) + \left(\frac{\pi x}{4} \right)$

$$= \frac{\pi}{2} - 3 + 2 + 7$$

$$= \frac{3\pi}{2} - 1$$

(ii) $A = \int_{-3}^5 h(x) dx = \frac{\pi}{2} + 3 + 2$

$$= \left(\frac{3\pi}{2} + 5 \right)$$

